Reinforcement Learning

Lecture 9: Model-based methods

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Lecture overview

Lecture covers chapter 8 in Sutton & Barto [1] and examples from David Silver [2]

1. **Model-based reinforcement learning**
   - taxonomy
   - overview
   - the simulation cycle
   - characteristics
2. **Integrated learning and planning**
   - Dyna-Q
   - characteristics
   - Monte Carlo tree search
3. **Model-based policy optimization**
   - Model ensembles
   - Model rollouts

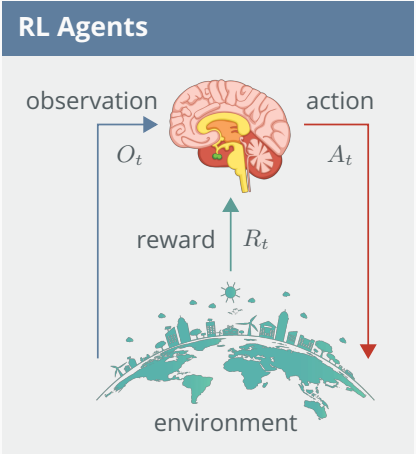
Model-based RL taxonomy

Taxonomy of reinforcement learning algorithms

- RL algorithms
  - Model-free
    - Policy optimisation
      - DPG
      - A3C
      - PPO
      - TD3
      - ...
    - Q-learning
      - DQN
      - Double DQN
      - Noisy DQN
      - TRPO
      - ...
  - Model-based
    - Learn model
      - World Models
      - I2A
      - MBMF
      - MBVE
      - ...
    - Given model
      - AlphaZero
      - ExIt
      - ...

This figure does not capture overlap, for example between policy optimsiation and Q-learning algorithms

Model-based RL overview



In **model-free** RL:

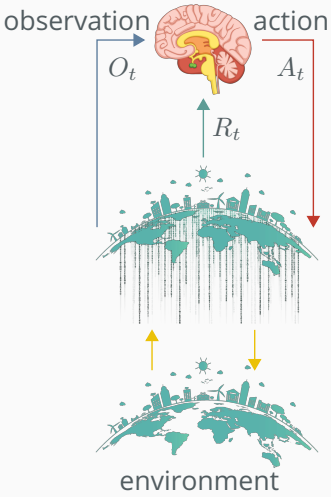
- No model
- **Learn** the value function $q(s, a)$ and/or the policy $\pi(a|s)$ from experience

In **model-based** RL:

- Learn the model from experience
- **Plan** the value function and/or the policy from the model

Figure based on [2, 1]

Model-based RL the simulation cycle



Model-based RL cycle:

- The agent experiences the real environment
- We learn a model to predict what the real environment does (when you take an action)
- We then use this simulated model to plan
- This allows us to estimate the value function and/or policy without directly interacting with the real environment
- But we use this policy to take real actions again

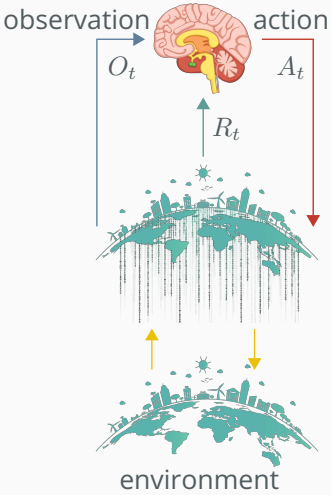
Model-based RL characteristics

Model-based RL advantages:

- The model may be much simpler than a value function
- Can be learnt by supervised learning
- Can reason about model uncertainty
- Can provide additional information (e.g. probabilities)
- Can change the reward function but keep the model

Model-based RL disadvantages:

- Another component that may introduce errors
- We can only be as good as our model
- Execution requires planning



Model-based RL definition

Definition: model

A model $\mathcal{M} = \langle \mathcal{P}_\eta, \mathcal{R}_\eta \rangle$ is a parameterised η representation of an MDP: $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R} \rangle$. It approximates state transitions $\mathcal{P}_\eta \approx \mathcal{P}$ and rewards $\mathcal{R}_\eta \approx \mathcal{R}$, learning a distribution over the next states and rewards:

$$S_{t+1} \sim \mathcal{P}(S_{t+1}|S_t, A_t)$$
$$R_{t+1} = \mathcal{R}(R_{t+1}|S_t, A_t, S_{t+1}),$$

which typically are conditionally independent of each other:

$$\begin{aligned} P(S_{t+1}, R_{t+1}|S_t, A_t) \\ &= P(S_{t+1}|S_t, A_t) P(R_{t+1}|S_t, A_t, S_{t+1}) \\ &:= P(S_{t+1}|S_t, A_t) P(R_{t+1}|S_t, A_t) \end{aligned}$$

Example: environment model

Learning the model

We learn the model $\mathcal{M}_\eta$ from experience $\{S_1, A_1, R_2, ..., S_T\}$ using **supervised learning**.

- We receive a stream of actual experiences
- This gives us a dataset:

$$S_1, A_1 \rightarrow R_2, S_2$$
$$S_2, A_2 \rightarrow R_3, S_3$$
$$...$$

- $s, a \rightarrow r$ is a regression problem
- $s, a \rightarrow s'$ is a density estimation problem

Integrated learning simulated and real experience

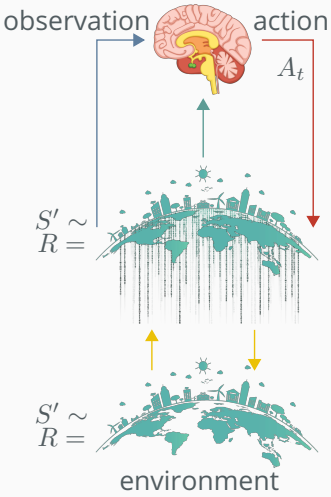
Experience can be simulated and real

Simulated experience sampled from $\mathcal{M}_\eta$

$$S' \sim \mathcal{P}_\eta(S'|S,A)$$
$$R = \mathcal{R}_\eta(R|S,A)$$

Real experience sampled from the true MDP

$$S' \sim \mathcal{P}^a_{s,s'}$$
$$R = \mathcal{R}^a_s$$

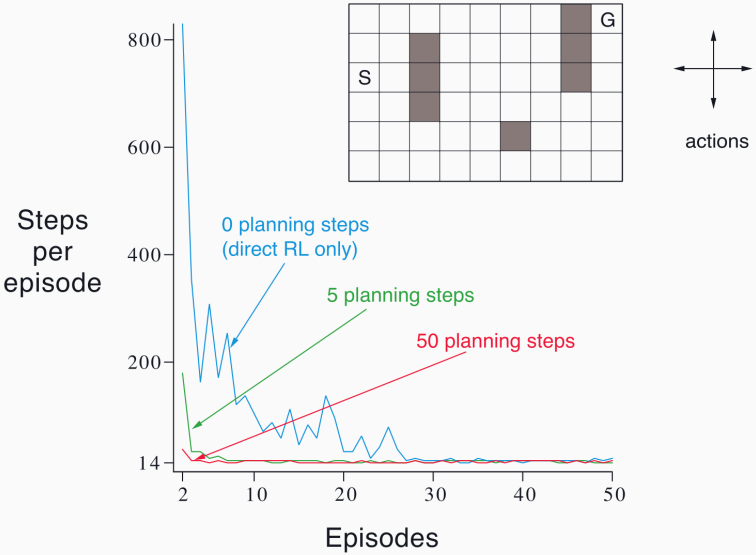


Model-based RL Dyna-Q

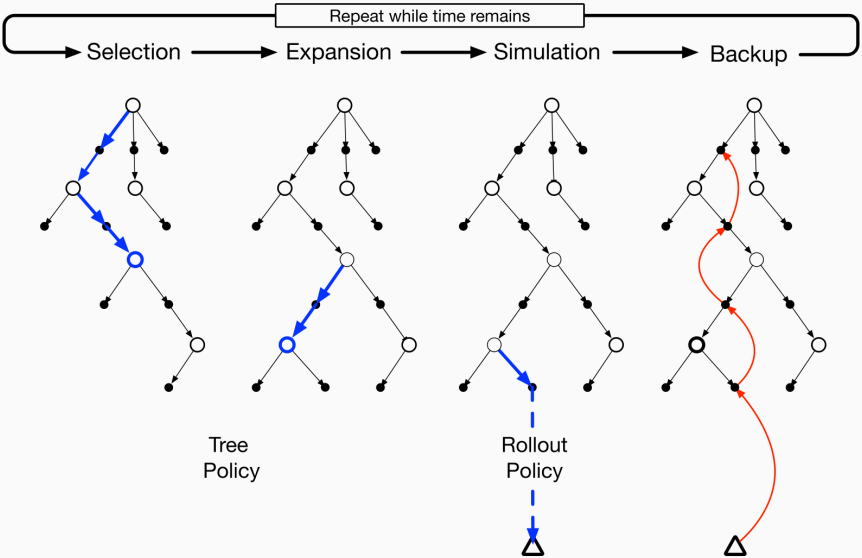
Algorithm: Dyna-Q [3, 4]

```
initialise Q(s,a) and model M(s,a) for all s ∈ S and a ∈ A(s)
while True:
  s ← current (nonterminal) state
  a ← ε−greedy(s,Q)
  r,s' ← env.step(s,a)
  Q(s,a) ← Q(s,a) + α(r + γ max_â Q(s',â) − Q(s,a))
  M(s,a) ← r,s' (assuming deterministic environment)
  for i in range(n):
    s ← random previously observed state
    a ← random action previously taken in s
    r,s' ← M(s,a)
    Q(s,a) ← Q(s,a) + α(r + γ max_â Q(s',â) − Q(s,a))
```

Model-based RL Dyna-Q characteristics



Model-based RL Monte Carlo tree search (MCTS)



Algorithm Monte Carlo tree search (MCTS)

Selection	Sample from ***tree policy*** (e.g. UCB1) until reaching leaf node
Expansion	Sample transition to expand the tree with new leaf node
Simulation	Sample from ***rollout policy*** (e.g. learned/fixed heuristic or random) until done
Backup	Compute return and update estimates along path

Revisiting the same state

Essentially does not happen:

- **Complex environments:** many states
- **Partially observability:** need history
- **Go/Chess:** game over (after 3x in chess)

Estimates per node

- **State-action values/policy**
  $Q(s, a) \to \pi(a \mid s) \approx \max Q(s, a)$
- **Transition function/model** $p(s' \mid s, a)$
- **Can use function approximation!**

Model-ensemble policy optimization ensembles

Algorithm Model-ensemble trust-region policy optimization (ME-TRPO) [5]

Initialize a policy π_θ and all models $\hat{f}_{\phi_1}, \hat{f}_{\phi_2}, ..., \hat{f}_{\phi_K}$.
Initialize an empty dataset $\mathcal{D}$.
repeat
    Collect samples from the real system f using π_θ and add them to D.
    Train all models using $\mathcal{D}$.
    repeat ▷ Optimize π_θ using all models.
        Collect fictitious samples from $\{\hat{f}_{\phi_i}\}_{i=1}^{K}$ using π_θ.
        Update the policy on the fictitious samples.
        Estimate the performances $\hat{\eta}(\theta; \phi_i)$ for $i = 1, ..., K$.
    until the performances stop improving.
until the policy performs well in real environment f.

Model-based policy optimization rollouts

Algorithm Model-based policy optimization (MBPO) [6]

Initialize policy π_ϕ, predictive model p_θ, environment dataset $\mathcal{D}_{\text{env}}$, model dataset $\mathcal{D}_{\text{model}}$
for N epochs **do**
  Train model p_θ on $\mathcal{D}_{\text{env}}$ via maximum likelihood
  for E steps **do**
    Take action in environment according to π_ϕ; add to $\mathcal{D}_{\text{env}}$
    for M model rollouts **do**
      Sample s_t uniformly from $\mathcal{D}_{\text{env}}$
      Perform k-step model rollout starting from s_t using policy π_ϕ; add to $\mathcal{D}_{\text{model}}$
    for G gradient updates **do**
      Update policy parameters on model data: $\phi \leftarrow \phi - \lambda_\pi \hat{\nabla}_\phi J_\pi(\phi, \mathcal{D}_{\text{model}})$

Take Away Points

Summary

In summary, model-based methods:

- are trained with supervised learning methods
- allow for planning ahead without interacting with the environment
- can be very data efficient by generating simulated experience
- but the values and policies can only be as good as the model
- they can be combined with model-free methods
- allow for changing the reward function/task without relearning the model

References I

[1] Richard S Sutton and Andrew G Barto. Reinforcement learning: An introduction (second edition). Available online. MIT press, 2018.

[2] David Silver. Reinforcement Learning lectures. https://www.davidsilver.uk/teaching/. 2015.

[3] Richard S Sutton. "Integrated architectures for learning, planning, and reacting based on approximating dynamic programming". In: Machine learning proceedings 1990. Elsevier, 1990, pp. 216–224.

[4] Baolin Peng et al. "Deep Dyna-Q: Integrating planning for task-completion dialogue policy learning". In: arXiv preprint arXiv:1801.06176 (2018).

[5] Thanard Kurutach et al. "Model-Ensemble Trust-Region Policy Optimization". In: International Conference on Learning Representations. 2018. URL: https://openreview.net/forum?id=SJJinbWRZ.

References II

[6] Michael Janner et al. “When to trust your model: Model-based policy optimization”. In: arXiv preprint arXiv:1906.08253 (2019).